



A Discrete Adjoint from the Run Record

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We want the derivative of a measure of a model's result with respect to a parameter the model depends on: here, the strength of a fault. A model in Underworld3 is a Python program, and there is no model document to differentiate. There is instead the residual of each implicit solve, which is a SymPy expression the solver assembled from what the user wrote. This note sets out how that residual gives an exact discrete adjoint for one solve, and uses it on a listric fault whose friction varies down dip.

1. AN IMPLICIT SOLVE IS A RESIDUAL

Every solver assembles its residual from two templates, F_0 and F_1 , in the unknown u and whatever fields and named quantities the user wrote into them:

$$R(u; m) = \int_{\Omega} F_0(u, \nabla u, m) \phi + F_1(u, \nabla u, m) \cdot \nabla \phi = 0. \quad (1)$$

The SNES that solves it assembles the Jacobian $K = \partial R / \partial u$ from the symbolic derivative of (1), and the same derivative can be taken with respect to any named quantity m in the residual. Nothing is linearised by hand and nothing is differenced.

2. THE ADJOINT OF ONE SOLVE

For a misfit $J(u)$ of the solution, the derivative with respect to m is

$$K^T \mu = -\frac{\partial J}{\partial u}, \quad \frac{dJ}{dm} = \frac{\partial J}{\partial m} + \mu^T \frac{\partial R}{\partial m}. \quad (2)$$

In the library this is one call, `solver.gradient(misfit, parameters=...)`, made of three steps. `misfit_duals` assembles $\partial J / \partial u$ as a load, $\int g_0 \phi_j + g_1 \cdot \nabla \phi_j$, where g_0 and g_1 are the symbolic derivatives of the misfit integrand with respect to u and ∇u . A misfit on a stress reads the velocity through its gradient, and the gradient part of the load is what carries it; there is no integration by parts and no boundary term. `adjoint_solve` assembles K^T rather than transposing K . PETSc assembles a Jacobian block from four pointwise kernels, the derivatives of F_0 and F_1 with respect to u and ∇u , and the transposed bilinear form has the same four with trial and test exchanged: the two diagonal kernels transposed on their paired indices and the two mixed ones swapped. The solver registers that kernel set, lets the SNES assemble, and solves the result with its own Krylov

solver and preconditioner; on the composite velocity–pressure system of Stokes the (u, p) and (p, u) blocks exchange places. Nothing is differentiated again, and the assembled operator agrees with the explicit transpose to 10^{-16} . sensitivity forms $\mu^T \partial R / \partial m$ by differentiating the residual with respect to m , expanding every other named expression first: the residual holds the constitutive model’s own symbol for the parameter, and the derivative of a value substituted too early is zero.

Whether the forward solve reached its state by Newton or by Picard does not enter. The converged state is the same, $\partial R / \partial u$ is a function of that state, and the residual is symbolic, so the consistent tangent is assembled for the transpose whichever tangent the iteration used.

3. A LISTRIC FAULT OF UNKNOWN FRICTION

The fault is listric: a flat décollement at depth that steepens on a circular ramp to a dip of 60° where it reaches the surface, in a box 20 km wide and 10 km deep meshed at 24 cells across the depth, under horizontal shortening from both sides and a gravity load, with a no-slip base and a free top, so the surface velocity is the uplift rate. The notebook states the problem in kilometres, pascal seconds and millimetres per year, and the solver works in units of the depth, the bulk viscosity of 10^{21} Pa s and the convergence rate of 8.4 mm/yr, so a unit of stress is 26 MPa and the lithostatic pressure at the base is ten. The numbers in this note are in those units. The fault is not cut into the mesh. It is a weak plane in a transversely isotropic viscosity: the bulk viscosity is $\eta_0 = 1$, the director is the normal at the nearest point of the fault, vertical on the flat and radial on the ramp, and the plane yields at the Coulomb stress

$$\tau_y = C + \mu p, \quad \eta_1 = \frac{\eta_0 \tau_y}{\tau_y + 2\eta_0 \dot{\epsilon}_s}, \quad (3)$$

with p the pressure, $\dot{\epsilon}_s$ the shear strain rate resolved on the plane, and η_1 the plane’s viscosity, which tends to $\tau_y / 2\dot{\epsilon}_s$ where the plane slips and to η_0 where it does not. Outside a band around the fault $\eta_1 = \eta_0$. The friction coefficient μ takes a separate value on each of four segments, each a named expression in the residual: the flat, the lower and upper thirds of the ramp, and the third nearest the surface. The residual is nonlinear in the velocity through $\dot{\epsilon}_s$ and in the pressure through τ_y , so the forward solve is Newton and the transpose in (2) is of the consistent tangent, which carries both dependences.

The observations are the uplift rate v_y along the top surface and the shear stress $2\eta_0 \dot{\epsilon}_{xy}$ near five points in the bulk, taken from a run at the true coefficients (0.05, 0.15, 0.25, 0.4): a weak décollement and a ramp whose friction rises towards the surface. The misfit is the squared difference of the uplift rate integrated along the surface, plus the squared difference of the stress under a Gaussian weight around each point; its dual on the velocity is a facet load and a volume load. We invert for the logarithm of each coefficient, starting from 0.2 on every segment, with L-BFGS from SciPy taking the misfit and its gradient from the three calls above.

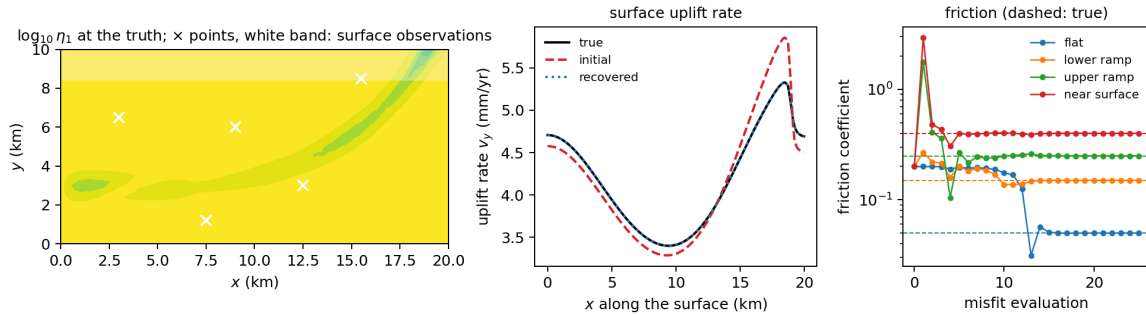


Figure 1: The plane’s viscosity at the true friction, with the stress points; the uplift rate at the true, initial and recovered coefficients; and the path each coefficient took. The true and recovered profiles coincide.

Before the inversion, the adjoint gradient at the starting guess is checked against central differences in each log-coefficient:

segment	adjoint	finite difference	ratio
flat	-1.00333×10^{-6}	-1.00333×10^{-6}	1.00000
lower ramp	-4.38768×10^{-4}	-4.38768×10^{-4}	1.00000
upper ramp	-3.23950×10^{-3}	-3.23950×10^{-3}	1.00000
near surface	-3.98863×10^{-3}	-3.98863×10^{-3}	1.00000

The inversion recovers $(0.05, 0.15, 0.25, 0.4)$ to machine precision in twenty-six misfit evaluations, each one Newton solve of six iterations from a cold start, one transposed solve, and four sensitivities. The flat’s sensitivity at the start is three orders of magnitude below the ramp’s: at a friction of 0.2 under seven units of pressure the décollement barely slips and the surface hardly sees it, and it is the last coefficient to move, between the eleventh and fifteenth evaluations. The notebook in `examples/`, also there as a script, runs the check in three minutes and the inversion in about ten on a laptop.

4. WHAT THE EXAMPLE DOES NOT SHOW

The fault’s position and shape are fixed. The observations are noise-free and the misfit carries no regularisation, so the twin experiment says the gradient is right and the four coefficients are identifiable from these observations, and nothing about how well they would be resolved from real ones. The same script with other observations recovers the same four coefficients to five figures: the orientation of the principal stress at the five interior points alone, in twenty-three evaluations, and the surface strain rate $\partial v_x / \partial x$ along the top alone, in thirty-eight. The flat’s sensitivity is a thousandth of the near-surface segment’s under every set, and it is always the last to arrive. Two observables were tried and dropped. The stress orientation on the free surface is a sign, since the shear strain rate vanishes where the traction does; what an inversion on it fitted was the discrete strain rate’s departure from that. And a Gaussian band beneath the surface, standing in for a surface integral, sees the volume it covers rather than the surface, and recovered coefficients that the true boundary integral does not resolve. With Gaussian noise on the observed velocity at a fraction of its rms, and PETSc TAO’s bounded quasi-Newton method as the driver with friction confined to $[0.005, 1]$, the coefficients fail in the reverse of the sensitivity order: at one percent the flat is ten percent low and the rest within two, at three percent the flat is a third low, and at ten percent the flat goes to the upper bound, a locked décollement, with the lower ramp halved to compensate. A bound hands an unresolved parameter the edge of its box. Written as a negative log posterior instead, the misfit as $\chi^2/2$ against its floor under the noise and a Gaussian prior of width one log unit about the start, the unresolved segments return to the prior and the resolved ones move:

at three percent noise the near-surface segment is found to five percent and the flat is not moved, and at ten percent only the near-surface segment moves, halfway. The fitted χ^2/N is 1.0001 in each case, so those are the answers the data hold. A run in time is a chain of solves like this one, differentiated in reverse order.